

Product Requirements Document (PRD): HybridChat

Version: 1.0
Date: April 1, 2026
Product Owner: Ashish Sharma
Target Platforms: Android, iOS, Web (Progressive Web App)

1. Overview

HybridChat is a secure, hybrid messaging app that enables **personal DMs, group chats, and broadcast channels** using both **Bluetooth** (for local, offline communication) and **internet** (for remote messaging). The app prioritizes **privacy, minimal server data storage, and end-to-end encryption**.

2. Objectives

- Enable seamless messaging in both online and offline environments.
 - Ensure privacy: **no message content stored on the server for Bluetooth messages**.
 - Support real-time availability detection (Bluetooth vs. internet).
 - Provide a secure, encrypted, and user-friendly experience.
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3. Features & Requirements

A. Core Features

Feature	Description	Priority
Personal DMs	1:1 encrypted chat between users.	P0
Group Chats	Multi-user encrypted conversations.	P0
Broadcast Channels	One-to-many messaging (e.g., announcements, alerts).	P1

Bluetooth Messaging	Direct device-to-device messaging (no server storage for content).	P0
Internet Messaging	Messages routed via server when devices are out of Bluetooth range.	P0
User Availability	UI indicates if a user is available via Bluetooth or internet.	P0
End-to-End Encryption	All messages encrypted; keys never leave user devices.	P0
Secure Authentication	JWT-based login with refresh tokens.	P0
Message Sync	Sync messages when switching between Bluetooth and internet modes.	P0
Push Notifications	Notify users of new messages (via Firebase Cloud Messaging or similar).	P1
Message Search	Search within chats (local only for Bluetooth messages).	P2
Media Sharing	Share images/files (encrypted, stored on server for internet mode).	P2

4. Tech Stack

A. Backend

- **Language:** Node.js (TypeScript)
- **Framework:** Express.js
- **Database:** MongoDB (Atlas for cloud)
- **Authentication:** JWT, bcrypt
- **Encryption:** libsodium (Signal Protocol) for E2EE, OpenSSL for key management
- **Real-time:** Socket.io (for internet mode)
- **API Documentation:** Swagger/OpenAPI

B. Frontend (Mobile)

- **Android:** Kotlin, Android Bluetooth API, Retrofit for REST, WebSockets
- **State Management:** Jetpack Compose (Android), Combine (iOS)
- **Encryption:** Native libsodium bindings

C. Frontend (Web - PWA)

- **Framework:** React.js (Next.js)
- **Styling:** Tailwind CSS
- **Bluetooth:** Web Bluetooth API (Chrome/Edge only)
- **Real-time:** Socket.io-client

D. DevOps & Infrastructure

- **Hosting:** AWS EC2 or Google Cloud Run
 - **CI/CD:** GitHub Actions
 - **Monitoring:** Sentry, Prometheus
 - **Containerization:** Docker, Kubernetes (if scaling)
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5. Database Schema (MongoDB)

A. Collections

1. Complete MongoDB Schema

A. Collections & Fields

MongoDB Collections (Detailed)

Collection	Fields (Key: Type)	Indexes	Notes
Users	<code>_id: ObjectId, username: String, email: String, passwordHash: String, publicKey: String, lastSeen: Date, bluetoothMac: String, isOnline: Boolean, deviceToken: String (for push)</code>	<code>email:1, username:1</code>	<code>publicKey</code> used for E2E encryption key exchange.

Messages	_id: ObjectId, senderId: ObjectId, receiverId: ObjectId, groupId: ObjectId, channelId: ObjectId, content: String, timestamp: Date, mode: String ("bluetooth"/"internet"), isEncrypted: Boolean, metadata: Object	senderId:1, receiverId:1, timestamp:1	content is encrypted; metadata for Bluetooth messages only.
Groups	_id: ObjectId, name: String, adminId: ObjectId, members: [ObjectId], createdAt: Date, groupPublicKey: String	name:1, adminId:1	groupPublicKey for group message encryption.
Channels	_id: ObjectId, name: String, adminId: ObjectId, subscribers: [ObjectId], createdAt: Date, isPublic: Boolean	name:1	Broadcast channels can be public or private.
BluetoothSessions	_id: ObjectId, userId1: ObjectId, userId2: ObjectId, lastConnected: Date, sessionKey: String, metadata: Object	userId1:1, userId2:1	sessionKey for Bluetooth message encryption.
Tokens	_id: ObjectId, userId: ObjectId, token: String, expiresAt: Date, createdAt: Date	userId:1, token:1	JWT refresh tokens.

B. Indexes

- Users: email:1 (unique), username:1 (unique)
 - Messages: senderId:1, receiverId:1, timestamp:1
 - Groups: name:1, adminId:1
 - Channels: name:1
 - BluetoothSessions: userId1:1, userId2:1
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6. API Routes

A. Authentication

Endpoint	Method	Description	Auth Required	Request Body	Response
/auth/register	POST	Register a new user.	No	{username, email, password, publicKey, bluetoothMac}	{user, token}
/auth/login	POST	Login and return JWT.	No	{email, password}	{user, token}
/auth/refresh	POST	Refresh JWT token.	Yes	{token}	{token}
/auth/logout	POST	Invalidate JWT.	Yes		{success: true}

B. Messaging

Endpoint	Method	Description	Auth Required	Request Body	Response
/messages/send	POST	Send a message (internet).	Yes	{receiverId, groupId, channelId,	{message, success: true}

					content, mode}	
/messages/sync	GET	Sync messages for a user.	Yes			{messages: [Message]}
/messages/bluetooth	POST	Log Bluetooth message metadata.	Yes	{receiverId, messageId, timestamp, metadata}		{success: true}
/messages/nearby	GET	List Bluetooth-available users.	Yes			{users: [User]}

C. Groups & Channels

Endpoint	Method	Description	Auth Required	Request Body	Response
/groups	POST	Create a new group.	Yes	{name, members: [userId]}	{group}
/groups/:id/messages	GET	Get group messages.	Yes		{messages: [Message]}
/channels	POST	Create a new channel.	Yes	{name, isPublic}	{channel}
/channels/:id/subscribe	POST	Subscribe to a channel.	Yes		{channel}
/channels/:id/messages	GET	Get channel messages.	Yes		{messages: [Message]}

7. Security & Encryption

A. Authentication

- **JWT:** Tokens expire in 24 hours; refresh tokens last 30 days.
- **Password Hashing:** bcrypt (12 rounds).
- **Rate Limiting:** 100 requests/minute/IP.

B. End-to-End Encryption

- **Key Exchange:** Users exchange public keys during registration/first contact.
- **Message Encryption:**
 - **1:1 Chats:** AES-256; key derived from sender's private key + receiver's public key.
 - **Group Chats:** Group admin generates **groupPublicKey**; all members encrypt with this key.
 - **Bluetooth:** Session key per Bluetooth session (stored in **BluetoothSessions**).
- **Libraries:** libsinal (Signal Protocol) for key management.

C. Data Privacy

- **Bluetooth Messages:** Only metadata stored on server.
 - **Internet Messages:** Encrypted content stored; decryption keys never leave devices.
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8. Bluetooth Integration

- **Scanning:** App scans for Bluetooth devices every 30 seconds.
 - **Matching:** Compares **bluetoothMac** with registered users.
 - **Direct Transfer:** Uses RFCOMM sockets; no server involvement.
 - **Fallback:** Auto-switches to internet mode if Bluetooth fails.
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9. UI/UX Requirements

- **User Availability:** Icons/colors indicate if a user is reachable via Bluetooth or internet.
- **Message Status:** Read receipts, delivery status (sent/delivered/read).
- **Encryption Indicators:** Padlock icon for encrypted messages.
- **Offline Mode:** Clear UI for Bluetooth-only messaging.

10. Success Metrics

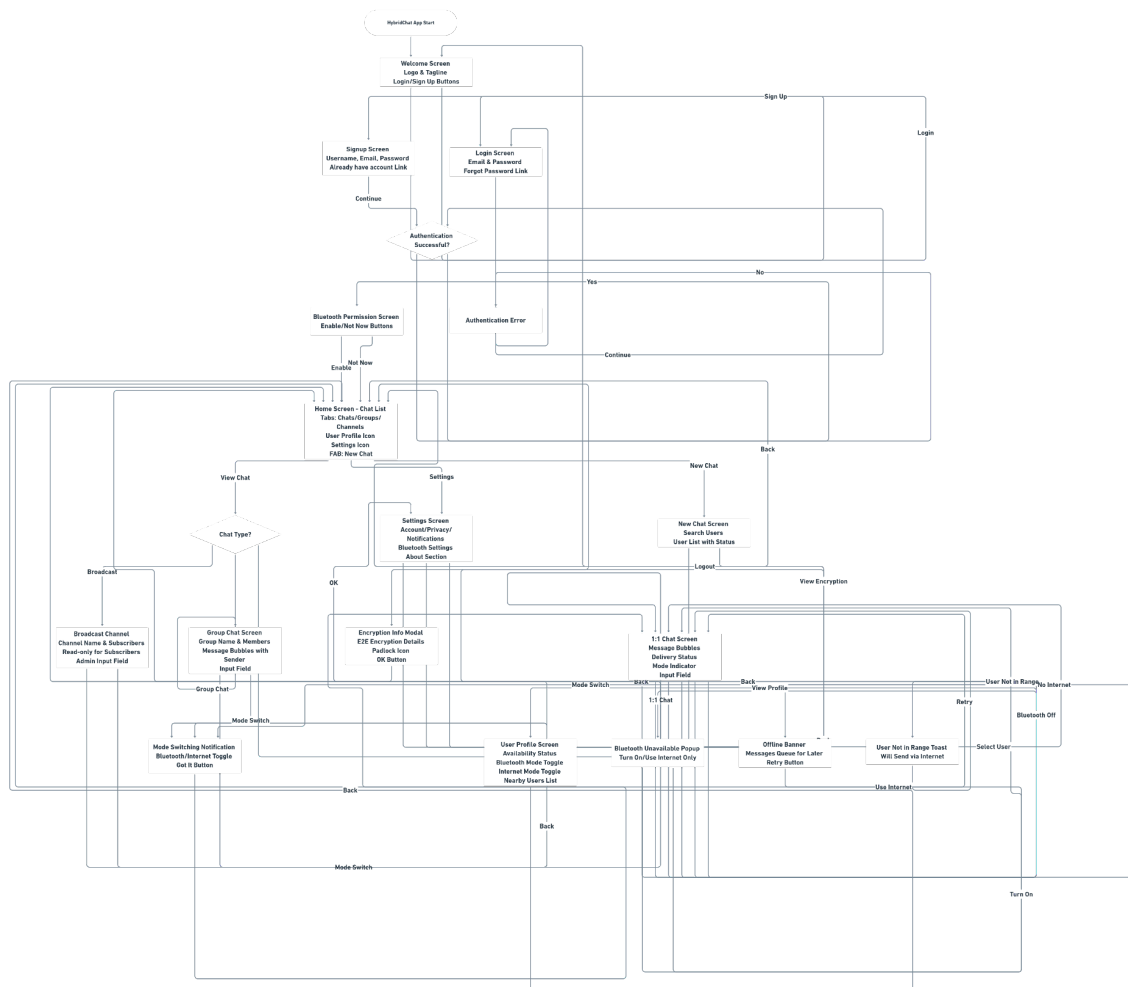
- **Adoption:** % of messages sent via Bluetooth vs. internet.
 - **Performance:** Message delivery latency (target: <1s for Bluetooth, <2s for internet).
 - **Security:** 0 breaches or unauthorized access incidents.
 - **User Satisfaction:** CSAT score > 4.5/5.
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11. Risks & Mitigations

Risk	Mitigation
Bluetooth reliability issues	Implement auto-fallback to internet mode.
Key management vulnerabilities	Use libsodium; regular security audits.
Server scalability	Use MongoDB Atlas; optimize indexes and queries.
Battery drain	Optimize Bluetooth scanning interval; use low-energy modes.

12. Roadmap

Phase	Timeline	Deliverables
1	Week 1-4	Backend API, MongoDB schema, authentication, basic messaging
2	Week 5-8	Bluetooth module, E2EE, group/channel features
3	Week 9-12	UI/UX, push notifications, testing, bug fixes
4	Week 13-16	Beta release, user feedback, final polish



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flowchart TD

HybridChat_App_Start_8d306eec([HybridChat App Start]) --> Welcome_Screen_Logo__f5c6020b["Welcome Screen
Logo & Tagline
Login/Sign Up Buttons"]

Welcome_Screen_Logo__f5c6020b -->|Sign Up| Signup_Screen_Userna_889c7309[Signup Screen
Username, Email, Password
Already have account Link]

Welcome_Screen_Logo__f5c6020b -->|Login| Login_Screen_Email_P_b6924758[Login Screen
Email & Password
Forgot Password Link]

Signup_Screen_Userna_889c7309 -->|Continue| Authentication_Succe_128cec73{Authentication
Successful?}

Login_Screen_Email_P_b6924758 -->|Continue| Authentication_Succe_128cec73

Authentication_Succe_128cec73 -->|Yes| Bluetooth_Permission_2a2cc1ae["Bluetooth Permission Screen
Enable/Not Now Buttons"]

Authentication_Succe_128cec73 -->|No| Authentication_Error_765b045b[Authentication Error]

Bluetooth_Permission_2a2cc1ae -->|Enable| Home_Screen_Chat_Lis_8e6a0624["Home Screen - Chat List
Tabs: Chats/Groups/Channels
User Profile Icon
Settings Icon
FAB: New Chat"]

Bluetooth_Permission_2a2cc1ae -->|Not Now| Home_Screen_Chat_Lis_8e6a0624

Authentication_Error_765b045b --> Login_Screen_Email_P_b6924758

Home_Screen_Chat_Lis_8e6a0624 -->|View Chat| Chat_Type_72d63ce7{Chat Type?}

Home_Screen_Chat_Lis_8e6a0624 -->|Settings| Settings_Screen_Acco_fcabf4a4["Settings Screen
Account/Privacy/Notifications
Bluetooth Settings
About Section"]

Home_Screen_Chat_Lis_8e6a0624 -->|New Chat| New_Chat_Screen_Sear_5d902cf0[New Chat Screen
Search Users
User List with Status]

Chat_Type_72d63ce7 -->|Broadcast| Broadcast_Channel_Ch_5cb1f447[Broadcast Channel
Channel Name & Subscribers
Read-only for Subscribers
Admin Input Field]

Chat_Type_72d63ce7 -->|Group Chat| Group_Chat_Screen_Gr_88b778be[Group Chat Screen
Group Name & Members
Message Bubbles with Sender
Input Field]

Chat_Type_72d63ce7 -->|1:1 Chat| 1_1_Chat_Screen_Mess_a9e3d5fd[1:1 Chat Screen
Message Bubbles
Delivery Status
Mode Indicator
Input Field]

Settings_Screen_Acco_fcabf4a4 -->|Logout| Welcome_Screen_Logo__f5c6020b

Settings_Screen_Acco_fcabf4a4 -->|Back| Home_Screen_Chat_Lis_8e6a0624

Settings_Screen_Acco_fcabf4a4 -->|View Encryption| Encryption_Info_Moda_d5e4600c[Encryption Info Modal
E2E Encryption Details
Padlock Icon
OK Button]

New_Chat_Screen_Sear_5d902cf0 -->|Back| Home_Screen_Chat_Lis_8e6a0624

New_Chat_Screen_Sear_5d902cf0 -->|Select User| 1_1_Chat_Screen_Mess_a9e3d5fd

Broadcast_Channel_Ch_5cb1f447 -->|Back| Home_Screen_Chat_Lis_8e6a0624

Broadcast_Channel_Ch_5cb1f447 -->|Mode Switch| Mode_Switching_Notif_7c9bb066["Mode Switching Notification
Bluetooth/Internet Toggle
Got It Button"]

Group_Chat_Screen_Gr_88b778be -->|Back| Home_Screen_Chat_Lis_8e6a0624

Group_Chat_Screen_Gr_88b778be -->|Mode Switch| Mode_Switching_Notif_7c9bb066

Encryption_Info_Moda_d5e4600c -->|OK| Settings_Screen_Acco_fcabf4a4

1_1_Chat_Screen_Mess_a9e3d5fd -->|Back| Home_Screen_Chat_Lis_8e6a0624

1_1_Chat_Screen_Mess_a9e3d5fd -->|Mode Switch| Mode_Switching_Notif_7c9bb066

1_1_Chat_Screen_Mess_a9e3d5fd -->|View Profile| User_Profile_Screen__a35862ca[User Profile Screen
Availability Status
Bluetooth Mode Toggle
Internet Mode Toggle
Nearby Users List]

1_1_Chat_Screen_Mess_a9e3d5fd -->|Bluetooth Off| Bluetooth_Unavailabl_8e2fb3fe["Bluetooth Unavailable Popup
Turn On/Use Internet Only"]

1_1_Chat_Screen_Mess_a9e3d5fd -->|No Internet| Offline_Banner_Messa_1b63fed4[Offline Banner
Messages Queue for Later
Retry Button]

1_1_Chat_Screen_Mess_a9e3d5fd -->|User Not in Range| User_Not_in_Range_To_4834b9ac[User Not in Range Toast
Will Send via Internet]

Mode_Switching_Notif_7c9bb066 --> Home_Screen_Chat_Lis_8e6a0624

User_Profile_Screen__a35862ca -->|Back| Home_Screen_Chat_Lis_8e6a0624

Bluetooth_Unavailabl_8e2fb3fe -->|Turn On| 1_1_Chat_Screen_Mess_a9e3d5fd

Bluetooth_Unavailabl_8e2fb3fe -->|Use Internet| 1_1_Chat_Screen_Mess_a9e3d5fd

Offline_Banner_Messa_1b63fed4 -->|Retry| 1_1_Chat_Screen_Mess_a9e3d5fd

User_Not_in_Range_To_4834b9ac --> 1_1_Chat_Screen_Mess_a9e3d5fd

classDef style0 stroke:#C3CFD9,fill:#FFFFFF

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1_1_Chat_Screen_Mess_a9e3d5fd,Authentication_Error_765b045b,Authentication_Succe_128cec73,Bluetooth_Permission_2a2cc1ae,Bluetooth_Unavailabl_8e2fb3fe,Broadcast_Channel_Ch_5cb1f447,Chat_Type_72d63ce7,Encryption_Info_Moda_d5e4600c,Group_Chat_Screen_Gr_88b778be,Home_Screen_Chat_Lis_8e6a0624,HybridChat_App_Start_8d306eec,Login_Screen_Email_P_b6924758,Mode_Switching_Notif_7c9bb066,New_Chat_Screen_Sear_5d902cf0,Offline_Banner_Messa_1b63fed4,Settings_Screen_Acco_fcabf4a4,Signup_Screen_Userna_889c7

309,User_Not_in_Range_To_4834b9ac,User_Profile_Screen__a35862ca,Welcome_Screen_Lo
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4,35,36,37 stroke:#788896